

Geospatial proposal readiness

Four contracts still block complete model placement. The local candidate supplies concrete answers for five former gaps.

64 tests passed. 37 cases and 168 queries per reader, with source data unchanged.

Headless OpenUSD, native OpenUSD and live OV resolve direct anchor origins. Full model placement remains stopped.

Concrete candidate answers

Subject	Proposed contract
Placement fields	Varying crs:position double3 has no fallback. crs:orientation quatd defaults to identity. crs:scale double3 defaults to (1,1,1). Explicit blocks remain unavailable.
Coordinate components	Easting/northing/up, longitude/latitude/height, or geocentric XYZ. Use declared WKT units and height reference. Scene upAxis does not relabel the tuple.
Orientation and scale	Model scale precedes orientation. Source bases are projected grid axes, geographic geodetic ENU or geocentric XYZ. The stage-axis adapter still needs specification.
WKT string normalization	OGC preferred spelling and delimiters, preserved quoted content and node order, W3C exact canonical decimal/integer spelling. Token identity differs from CRS equivalence.
Comparison and sampling	Separate operation accuracy, approximation error and reader agreement. Actual timeSamples and explicit timeCodesPerSecond record the sampling schedule.

Four contracts still required

Contract	What remains unspecified
G03 / Q3: project-adjustment frame	Axes, units, pivot and controlling composed context for ordinary project adjustments, including nested bindings and output changes. Post-placement order is settled.
G04 / Q6: stage and placement bases	Exact mapping between stage axes/distances, the proposed source placement bases and resolved result bases. Coordinate tuples do not supply that mapping.
G06 / Q11: measurement association	Authored coordinate roles/domains for USD properties and external assets, including multiple domains and conflicting external CRS metadata.
G07 / Q12: dependency declaration	Authored carrier and conservative assembly/export maintenance through references, unloaded payloads and composition, readable without traversal.

Question 9 separately concerns geographic scene geometry, frames and bounds. Geographic coordinate queries already belong to initial scope.

WKT string normalization

24 tests check lexical identity while preserving represented information.

WKT fragment	Before	Normalized
Numeric spelling	LENGTHUNIT["metre",1E0]	LENGTHUNIT["metre",1.0]
Exact decimal	6378137.12345678912345	6378137.12345678912345
Quoted content	REMARK["Keep spaces"]	REMARK["Keep spaces"]

Different display names retain different tokens. CRS equivalence requires a separate engine comparison.

Illustrative fragments from the tested lexical rules. Context-sensitive legacy UNIT aliases are unsupported.

Executed reader coverage

Control family	Cases per reader	Queries per reader	Expected failures
Binding and composition	12	19	6
Source placement values	12	12	6
Direct anchor-origin coordinates	13	137	5

Every reader checks the same authored scenes. Direct origins include France projections, Colorado site calibration and analytic ECEF conversions.

Expected failures include unsupported frame requests. Source data stays unchanged. PROJ is shared across readers.

Same France origin in each runtime

France point 0 serves as an illustrative model-anchor origin. Source: RGF93 v2b / CC49 with NGF-IGN69 height. Output: Lambert-93 with the same height reference. Every displayed component is in metres.

Component	Source CC49	Headless USD	Native USD	Live OV
Easting	1661099.0390	661101.1094	661101.1094	661101.1094
Northing	8180053.1170	6857834.2565	6857834.2565	6857834.2565
Height	37.2980	37.2980	37.2980	37.2980

All 59 France origins pass in both directions. Coordinates only, with no geometry or measurement-domain claim.

Source interpolation and export samples

A WGS84 anchor moves from longitude -1 degree to +1 degree on the equator at zero height. At time 5, Core linear source interpolation yields longitude zero before ECEF conversion. Each reader returns $X = 6378137$ m, $Y = 0$ m, $Z = 0$ m.

Time code	Source longitude (degrees)	Resolved ECEF X (m)
0	-1	6377165.578842
5	0	6378137.000000
10	1	6377165.578842

Interpolating only the converted endpoints misses the midpoint by 971.421158 m.

Fresh USD sampling record: keys 0, 5, 10 and explicit timeCodesPerSecond 48. This is not a resolved-scene export.

Postimplementation audit

The audit tightened the partial origin adapter to reject nested/descendant frame requests rather than applying an incomplete model. It also corrected geographic identity reporting to use exact component comparison instead of a length metric. These repairs change consumer support/reporting, add no scene fact and do not amend the proposal to match code. Source fixtures author only crs:wkt and the three candidate placement fields. Resolution rewrites no source xformOps, resetXformStack, definitions, bindings or sampling keys.

Full model frames, project-adjusted placement, geometry and bounds, Hydra/OV resolved geometry, measurement-associated data and conservative resolved-scene export depend on the four missing contracts. These partial tests do not certify near-unit quaternion validation policy or geographic-distance controls. Group adoption and geodetic accuracy remain distinct from passing execution.